



HUMAN ACTIVITY RECOGNITION FROM SMARTPHONE SENSORS

PRESENTED BY

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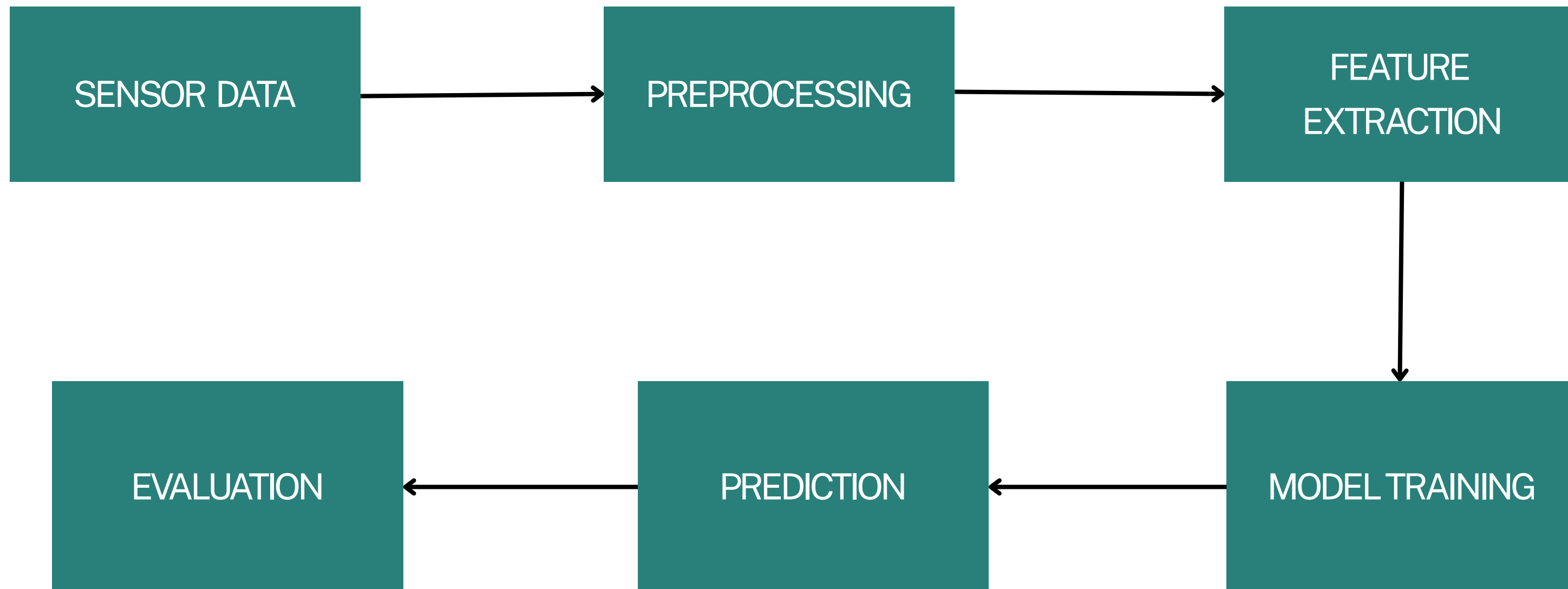
LANIYA MOHAN



PROBLEM STATEMENT

- Human Activity Recognition (HAR) using smartphone sensor data
- Classify activities: Walking, Sitting, Standing, Upstairs, Downstairs, Laying
- Uses accelerometer and gyroscope data
- Challenges: noisy data, similar activity patterns
- Applications: healthcare, fitness tracking, smart systems

Pipeline Diagram



DATASET OVERVIEW

- Dataset: UCI HAR Dataset
- ~10,000 samples collected from smartphones
- Sensors: Accelerometer (X, Y, Z), Gyroscope (X, Y, Z)
- 6 activity classes:
 - Walking, Walking Upstairs, Walking Downstairs
 - Sitting, Standing, Laying
- Dataset is balanced



DATA PREPROCESSING AND EDA

◆ PREPROCESSING

- Structured raw sensor data
- Cleaned and validated dataset
- Applied normalization
- Encoded activity labels

◆ Exploratory Analysis

- Studied activity distribution
- Analyzed sensor signal patterns
- Identified trends and variations
- Checked for imbalance and anomalies

FEATURE ENGINEERING APPROACH

Handcrafted
features (561)
extracted for SVM
and Random Forest

Includes time-
domain and
frequency-domain
features

LSTM uses raw time-
series input (128×6)

Automatic feature
learning in LSTM vs
manual extraction in
ML models

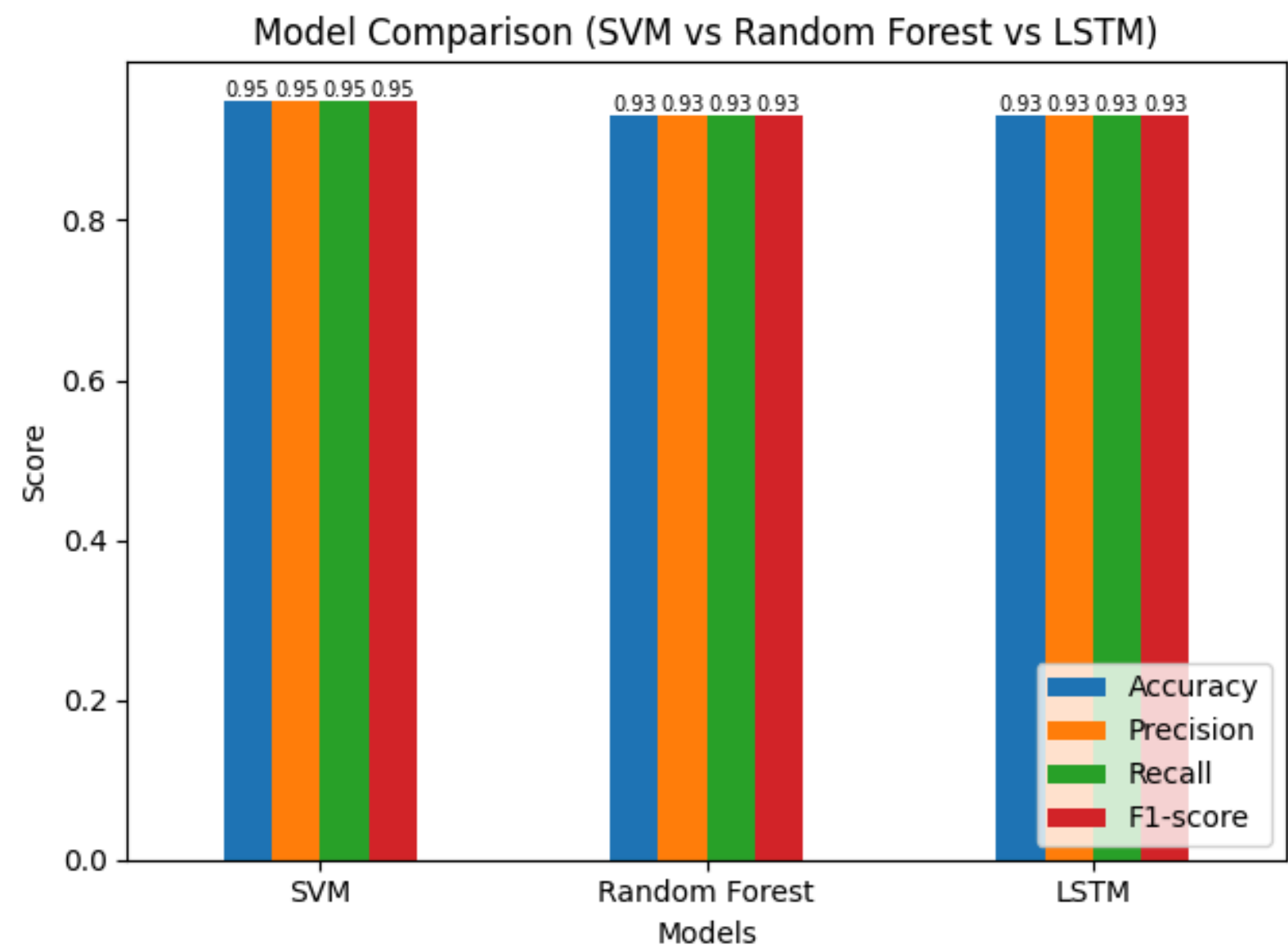
MODEL BUILDING AND TRAINING

- Trained three models:
 - SVM (Support Vector Machine)**
 - Random Forest**
 - LSTM (Deep Learning)**
- SVM → works well for high-dimensional data
- Random Forest → robust & reduces overfitting
- LSTM → captures temporal patterns
- Training done on processed dataset

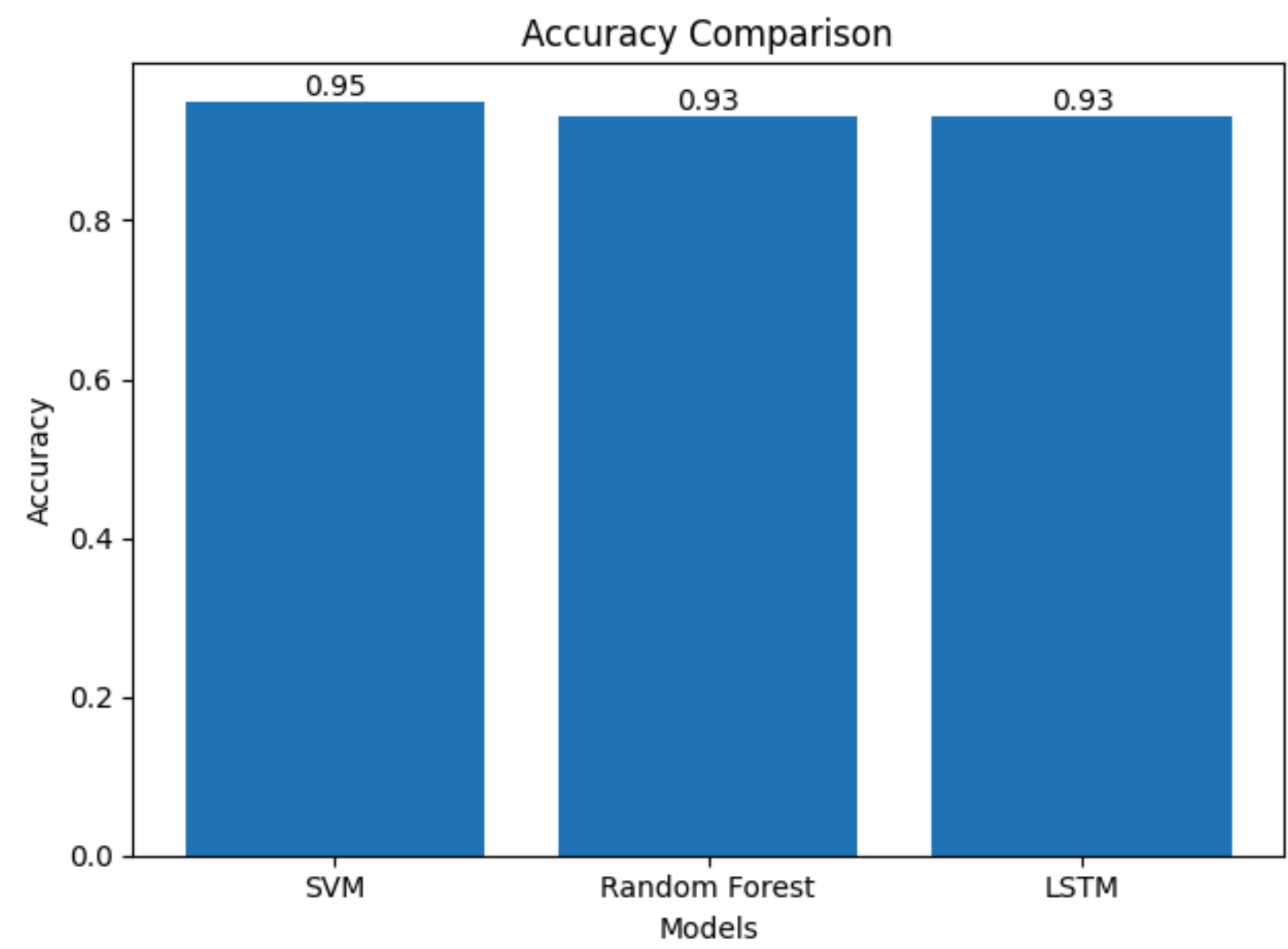


Results

MODELS	TRAINING ACCURACY	TEXT ACCURACY
SVM	~99%	~95%
Random Forest	~99%	~92%
LSTM	~98%	~92%

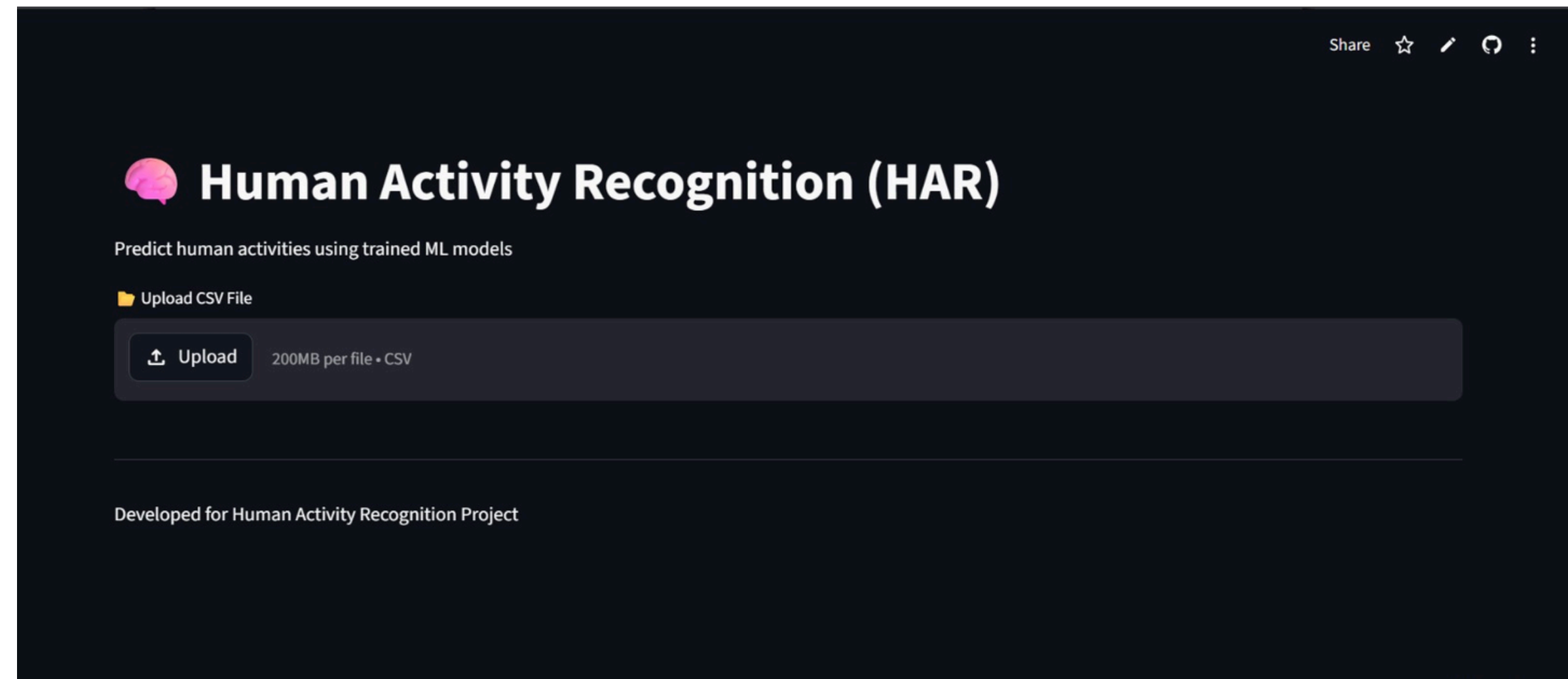
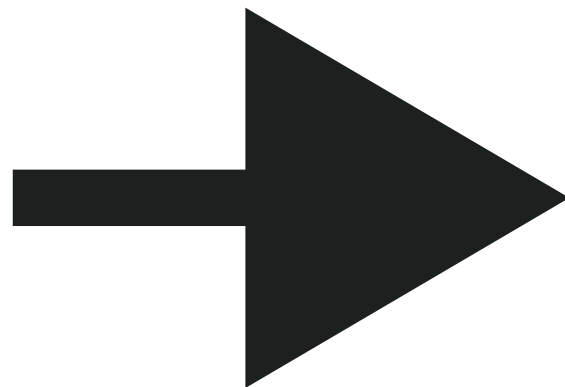


Accuracy Comparison



Application Screenshots

Home Interface



File Upload

Upload CSV File

X_test.csv

18.4MB

+

Prediction Mode

Choose mode

Single Row

Single Row

Random Row

Batch Prediction

Prediction: STANDING

Prediction

 **Prediction Mode**

Choose mode

Single Row

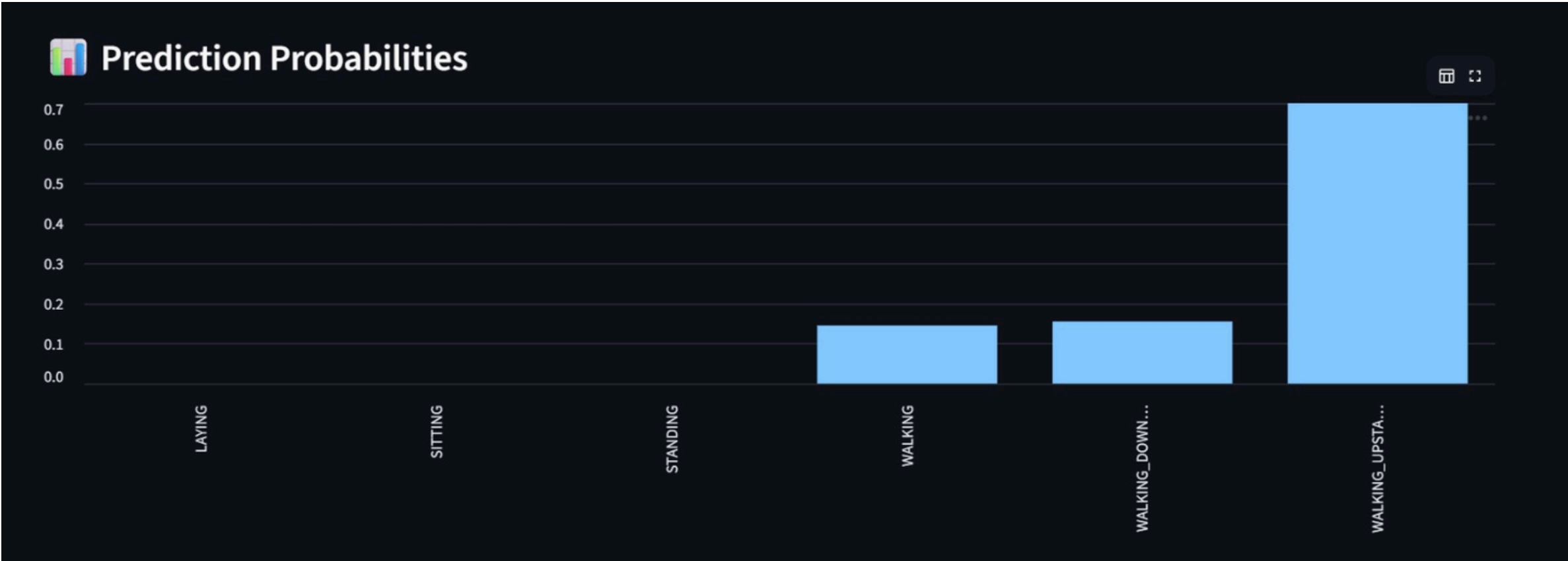
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Prediction: SITTING

Prediction Probabilities



Live Application Link

<https://humanactivityrecognitionproj1-ayeo2sbtfwgstmhxy5xtl.streamlit.app/>

Conclusion

- The model successfully classifies human activities using sensor-based features
- Random Forest achieved high performance with strong generalization
- The system effectively distinguishes between activities like walking, sitting, and standing
- Streamlit interface enables easy real-time prediction and visualization
- Feature-based approach proves effective for human activity recognition

Limitations

- High dimensional dataset (561 features) increases computational complexity
- Slight overfitting observed in some models (train accuracy > test accuracy)
- Model depends on pre-extracted features (no raw sensor processing)
- Limited to predefined activity classes only
- Performance may reduce in real-world noisy sensor environments
- No real-time streaming data processing implemented

**Thank you
very much!**